

KUBERNETES (K8s) COMMANDS CHEAT SHEET – FULL DEVOPS EDITION

1. Basic Cluster Commands

kubectl version

Show client + server K8s version.

kubectl cluster-info

See cluster master & DNS info.

kubectl get all

List all resources (pods, svc, deployments, etc.) in default namespace.

2. Working With Contexts

kubectl config get-contexts

List all contexts (clusters).

kubectl config use-context dev

Switch to another cluster.

kubectl config current-context

Show which cluster you are using.

3. Namespaces

kubectl get namespaces

List namespaces.

kubectl create namespace dev

Create namespace.

kubectl delete namespace dev

Delete namespace.

kubectl config set-context --current --namespace=dev

Set default namespace.

4. Pods

kubectl get pods

List pods.

kubectl get pods -o wide

Show pod IP, node, etc.

kubectl describe pod pod-name

Detailed pod info & events.

kubectl logs pod-name

View logs.

kubectl logs -f pod-name

Follow live logs.

kubectl exec -it pod-name -- bash

Enter pod terminal.

kubectl delete pod pod-name

Delete a pod.

5. Deployments

kubectl get deployments

List deployments.

kubectl create deployment web --image=nginx

Create deployment.

kubectl scale deployment web --replicas=5

Scale number of pods.

kubectl rollout status deployment/web

Check deployment rollout status.

kubectl rollout undo deployment/web

Rollback to previous version.

kubectl delete deployment web

Delete deployment.

6. ReplicaSets

kubectl get rs

List ReplicaSets.

kubectl describe rs rs-name

ReplicaSet details.

7. Services

kubectl get svc

List services.

kubectl expose deployment web --type=NodePort --port=80

Expose deployment as a service.

kubectl describe svc web

Service details.

kubectl get svc -o wide

See service cluster IP & ports.

8. ConfigMaps

```
kubectl create configmap app-config --from-literal=env=prod
```

Create ConfigMap.

```
kubectl create configmap myconfig --from-file=config.properties
```

Create from file.

```
kubectl get configmaps
```

List ConfigMaps.

```
kubectl describe configmap app-config
```

View config details.

9. Secrets

```
kubectl create secret generic db-secret --from-literal=password=1234
```

Create secret.

```
kubectl create secret generic tls-secret --from-file=server.crt --from-file=server.key
```

Create TLS secret.

```
kubectl get secrets
```

List secrets.

```
kubectl describe secret db-secret
```

Describe secret.

10. YAML Apply, Update & Delete

```
kubectl apply -f deployment.yaml
```

Apply or update resource.

kubectl delete -f deployment.yaml

Delete resource.

kubectl edit deployment web

Edit resource live in editor.

11. Nodes & Cluster Info

kubectl get nodes

List nodes.

kubectl get nodes -o wide

Node details (OS, internal IP).

kubectl describe node node1

Node details (taints, capacity).

kubectl drain node1 --ignore-daemonsets

Drain node safely.

kubectl cordon node1

Mark node unschedulable.

kubectl uncordon node1

Mark node schedulable.

12. Resource Usage

kubectl top pods

CPU & memory usage of pods.

kubectl top nodes

CPU & memory usage of nodes.

(Metrics server required)

13. Taints & Tolerations

Add taint:

```
kubectl taint nodes node1 key=value:NoSchedule
```

Remove taint:

```
kubectl taint nodes node1 key=value:NoSchedule-
```

Used for:

- ✓ Dedicate nodes
 - ✓ Restrict workloads
 - ✓ Isolation
-

14. Labels & Selectors

```
kubectl label pod web app=frontend
```

Add label.

```
kubectl get pods -l app=frontend
```

Filter with label.

```
kubectl label pod web app-
```

Remove label.

15. Port Forwarding

```
kubectl port-forward pod/mypod 8080:80
```

Access pod locally.

Common for debugging APIs.

16. Ingress

```
kubectl get ingress
```

List ingress rules.

kubectl describe ingress my-ingress

Ingress details.

17. StatefulSets

kubectl get statefulsets

List StatefulSets.

kubectl describe statefulset mysql

Stateful app details.

Used for:

Databases

Kafka

ElasticSearch

18. DaemonSets

kubectl get daemonsets

List DaemonSets.

Used for:

Logging agents

Monitoring agents

19. Jobs & CronJobs

kubectl create job myjob --image=busybox

Create Job.

**kubectl create cronjob backup --image=busybox --
schedule="*/5 * * * *"**

Create CronJob.

kubectl get jobs

List jobs.

kubectl get cronjobs

List CronJobs.

20. Pod Debugging

kubectl describe pod web

Check events.

kubectl logs pod --previous

Check logs of crashed container.

kubectl exec -it web -- sh

Debug inside pod.

kubectl get pod web -o yaml

View complete pod spec.

21. Troubleshooting Cluster

kubectl get events

Cluster-wide events.

kubectl get endpoints

Check service endpoints.

kubectl get componentstatus

Master component status (older K8s).

kubectl get networkpolicies

Check network restrictions.

22. Network Policies

kubectl get netpol

List policies.

kubectl describe netpol

Network policy details.

Used for:

Restrict pod-to-pod communication

Zero-trust networking

23. Storage

kubectl get pv

List persistent volumes.

kubectl get pvc

List persistent volume claims.

kubectl describe pv pv-name

PV details.

kubectl describe pvc pvc-name

PVC details.

24. Service Accounts & RBAC

kubectl get serviceaccounts

List service accounts.

kubectl create serviceaccount dev-sa

Create SA.

kubectl get clusterrole

List cluster roles.

kubectl get clusterrolebinding

List bindings.

Used for:

- ✓ Access control
 - ✓ Least privilege
 - ✓ Pod-to-AWS auth (IRSA)
-

25. Deleting Everything

kubectl delete all --all

Delete pods, svc, deployments in namespace.

kubectl delete namespace dev

Delete entire namespace.

26. Useful Shortcuts

kubectl get po

Short for pods.

kubectl get deploy

Short for deployments.

kubectl get svc

Short for service.

kubectl get ing

Short for ingress.

kubectl get cm

Short for configmap.

kubectl get no

Short for nodes.

27. Apply with Dry Run (very important)

kubectl apply -f app.yaml --dry-run=client

Check if YAML is valid without applying.

28. Debugging Node Issues

```
kubectl describe node node1
```

Check taints, disk pressure, memory pressure.

```
kubectl get nodes -o json | jq  
' .items[].status.conditions'
```

Detailed node conditions.

29. Extract Pod YAML

```
kubectl get pod web -o yaml > pod.yaml
```

Export pod definition.

Useful for reproducing or modifying pods.

30. Kubernetes API Access

```
kubectl proxy
```

Start API server proxy for debugging.

31. Advanced Pod Debugging

```
kubectl exec -it pod --container app -- sh
```

Exec into a specific container inside a multi-container pod.

```
kubectl logs pod -c app
```

View logs of a specific container.

```
kubectl logs pod --previous
```

Check logs of a container that crashed.

```
kubectl describe pod pod | grep -i error
```

Search for errors in pod events.

32. Ephemeral Debug Container (K8s 1.23+)

```
kubectl debug podname -it --image=busybox
```

Attach a temporary debug container.

```
kubectl debug node/node1 -it --image=busybox
```

Debug an entire node.

Used for:

- ✓ CrashLoopBackOff
 - ✓ ImagePullBackOff
 - ✓ Node-level debugging
-

33. ImagePull & Crash Issues

```
kubectl get pod pod -o  
jsonpath='{.status.containerStatuses[].state.waiting.reason}'
```

Get reason for failed pod.

Common issues:

- ImagePullBackOff
 - ErrImagePull
 - CrashLoopBackOff
-

34. Rollouts & History

```
kubectl rollout history deployment web
```

View deployment history.

```
kubectl rollout undo deployment web --to-revision=2
```

Rollback to a specific version.

```
kubectl rollout pause deployment web
```

Pause rollout (for debugging).

```
kubectl rollout resume deployment web
```

Resume rollout.

35. Advanced Scaling

kubectl scale deployment web --replicas=0

Scale down to zero.

kubectl scale statefulset mysql --replicas=3

Scale StatefulSet (ordered).

36. YAML Dry Run & Diff

kubectl diff -f app.yaml

Show changes before applying.

kubectl apply -f app.yaml --dry-run=server

Server-side validation without apply.

Useful for CI/CD pipelines.

37. Patching (Very Important for DevOps)

Patch resources inline:

kubectl patch deployment web -p '{"spec": {"replicas": 4}}'

Patch with strategic merge:

kubectl patch svc web --patch-file patch.yaml

Used for:

- ✓ Hotfixes
 - ✓ On-the-fly changes
 - ✓ CI/CD automation
-

38. Node Health & Troubleshooting

```
kubectl get nodes -o json | jq  
' .items[].status.conditions[] | select(.status=="False") '
```

Find unhealthy nodes.

```
kubectl describe node node1
```

Check disk pressure, memory pressure.

```
kubectl get pods --all-namespaces -o wide --field-selector spec.nodeName=node1
```

List all pods on a node.

39. Events & Clusterd-level Issues

```
kubectl get events
```

Show all events.

```
kubectl get events --sort-by='.lastTimestamp'
```

Sort events by time.

```
kubectl get --raw /metrics
```

Access kube-apiserver metrics.

40. Service Debugging

```
kubectl get endpoints web
```

Check if service has endpoints.

```
kubectl describe svc web
```

Check wrong targetPort / selector.

```
kubectl run test --image=busybox -it --rm -- wget web
```

Test service connectivity.

41. Ingress Debugging

kubectl get ingress

List ingress.

kubectl describe ingress web-ingress

Check rules & events.

kubectl get pods -n ingress-nginx

Check ingress controller health.

42. Network Policy Debugging

kubectl get netpol

List network policies.

kubectl describe netpol allow-app

View network rules.

Test connectivity:

`kubectl run test --rm -it --image=busybox -- wget http://pod-ip`

43. Persistent Storage Commands

kubectl get storageclass

List storage classes.

kubectl describe storageclass gp2

Details of storage class.

kubectl get pv

List persistent volumes.

kubectl get pvc

List persistent volume claims.

```
kubectl describe pvc my-pvc
```

PVC status.

44. Logs & Audit

```
kubectl logs pod --since=1h
```

Logs from the last 1 hour.

```
kubectl logs pod --tail=50
```

Show last 50 lines.

```
kubectl logs pod --timestamps
```

Show logs with timestamps.

45. Node-to-Pod Debugging

```
kubectl run debug --rm -it --image=busybox --command --  
ping <pod-ip>
```

Test pod connectivity.

```
kubectl get pod -o wide
```

Find pod IP.

46. Copy Files To & From Pod

```
kubectl cp pod:/app/logs ./logs
```

Copy from pod → local.

```
kubectl cp file.txt pod:/app/
```

Copy local → pod.

47. ConfigMaps & Secrets Debugging

```
kubectl get configmaps -o yaml
```

View ConfigMap values.

```
kubectl get secret db-secret -o yaml
```

Base64 encoded output.

Decode:

```
echo 'cGFzc3dvcmQ=' | base64 --decode
```

48. K8s Useful JSONPath Queries

Get pod image:

```
kubectl get pod web -o jsonpath='{.spec.containers[*].image}'
```

Get pod node:

```
kubectl get pod web -o jsonpath='{.spec.nodeName}'
```

49. Resource Quotas & Limits

```
kubectl get resourcequota
```

List quotas.

```
kubectl describe resourcequota rq1
```

Quota details.

50. LimitRanges

```
kubectl get limitrange
```

List pod limit ranges.

Used for:

- ✓ default CPU/memory
 - ✓ maximum/minimum allowed resources
-

51. Service Account with Pod

```
kubectl describe sa my-sa
```

Show token & secrets.

```
kubectl describe pod pod | grep ServiceAccount
```

Check which service account pod uses.

52. RBAC Debugging

Test user permission:

```
kubectl auth can-i get pods --as user1
```

Test namespace-specific permission:

```
kubectl auth can-i delete pods -n dev
```

53. Port & Connectivity Debugging (Must Know)

Test port:

```
kubectl run test --rm -it --image=busybox -- nc -zv web 80
```

DNS check:

```
kubectl run test --rm -it --image=busybox -- nslookup web
```

54. Horizontal Pod Autoscaling

```
kubectl autoscale deployment web --cpu-percent=50 --min=2 --max=10
```

Create autoscaler.

```
kubectl get hpa
```

List HPAs.

55. Cluster Autoscaler Debugging

```
kubectl -n kube-system get pods | grep autoscaler
```

Check autoscaler pod.

56. Advanced Resource Filtering (field selectors)

Get pods scheduled on a specific node:

```
kubectl get pods --field-selector spec.nodeName=node1
```

Get pods that failed:

```
kubectl get pods --field-selector=status.phase=Failed
```

Get pods that are not running:

```
kubectl get pods --field-selector=status.phase!=Running
```

Get pods created in last 5 minutes:

```
kubectl get pods --sort-by=.metadata.creationTimestamp
```

57. Advance JSON & YAML Output Formatting

Get only pod names:

```
kubectl get pods -o jsonpath='{.items[*].metadata.name}'
```

Get node internal IPs:

```
kubectl get nodes -o jsonpath='{.items[*].status.addresses[?(@.type=="InternalIP")].address}'
```

Get pod image names:

```
kubectl get pods -o jsonpath='{.items[*].spec.containers[*].image}'
```

58. Temporary BusyBox Pod (for debugging)

(one of the most used commands by DevOps engineers)

```
kubectl run tmp --rm -it --image=busybox -- sh
```

Used for:

- ✓ DNS testing
 - ✓ Connecting to services
 - ✓ Checking network restrictions
-

59. Check Pod Events Script

Helpful command to sort events:

```
kubectl get events --sort-by='.lastTimestamp'
```

Best for:

- ✓ CrashLoopBackOff
 - ✓ Pod scheduling failure
 - ✓ Image pull issues
-

60. Get Pod Environment Variables

```
kubectl exec -it <pod> -- printenv
```

Get env from container:

```
kubectl get pod <pod> -o jsonpath='{.spec.containers[*].env}'
```

61. Debugging Service DNS

Test DNS resolution:

```
kubectl run dns-test --rm -it --image=busybox -- nslookup svc-name
```

Test cluster DNS:

```
kubectl run dns-test --rm -it --image=busybox -- nslookup kubernetes.default
```

62. Debug Service Connectivity

Test port connectivity:

```
kubectl run test --rm -it --image=busybox -- nc -zv svc-name 80
```

Test via curl:

```
kubectl run test --rm -it --image=curlimages/curl -- curl http://svc-name
```

63. Debug Node Ports

Check nodeport access:

```
curl <node-ip>:<node-port>
```

64. Pod Security / User Permissions

Find which user the pod runs as:

```
kubectl get pod pod -o  
jsonpath='{.spec.containers[*].securityContext.runAsUser}'
```

65. Copy Kubernetes Manifest From Live Resource

Extract current running deployment YAML:

```
kubectl get deploy web -o yaml > web.yaml
```

Extract current configmap YAML:

```
kubectl get cm app-config -o yaml > configmap.yaml
```

66. Validating YAML

Client-side validation

```
kubectl apply -f app.yaml --dry-run=client
```

Server-side validation

```
kubectl apply -f app.yaml --dry-run=server
```

Lint YAML (if installed):

```
kubeval app.yaml
```

67. Kubernetes API Access (Raw)

View cluster components metrics:

```
kubectl get --raw /metrics
```

Access API paths:

```
kubectl get --raw /api
```

68. Node Disk / Memory Pressure Debugging

Check node conditions:

```
kubectl describe node node1 | grep -i pressure
```

Look for:

- DiskPressure
 - MemoryPressure
 - PIDPressure
-

69. Node Logs (Master & Worker Debugging)

Kubelet logs:

```
journalctl -u kubelet -f
```

Container runtime logs:

```
journalctl -u containerd -f
```

70. Restarting Pods Properly

Delete pod safely (deployment recreates it):

```
kubectl delete pod pod-name
```

Force delete stuck pod:

```
kubectl delete pod pod-name --force --grace-period=0
```

71. Restart Deployment (Without Editing)

```
kubectl rollout restart deployment web
```

72. Checking Cluster Authentication

Test if user can perform action:

```
kubectl auth can-i create pods
```

Test as specific user:

```
kubectl auth can-i delete pods --as bob
```

73. Get Logs From ALL Pods of a Deployment

```
kubectl logs -l app=web --all-containers=true
```

Labels must match deployment selector.

74. Debugging Network Policies

List policies:

```
kubectl get netpol
```

Connectivity test across namespaces:

```
kubectl run test --rm -it --image=busybox -- sh
```

Inside:

```
wget http://pod-ip
```

75. ConfigMap Reload Troubleshooting

Pods do NOT automatically reload ConfigMaps unless:

- ✓ Pod restarts
- ✓ Sidecar reloaders (e.g., Reloader, ConfigMap reloader)
- ✓ Using projected volumes

Check if mounted as volume:

```
kubectl get pod pod -o jsonpath='{.spec.volumes[*].configMap.name}'
```

76. Secret Decoding & Validation

Decode a secret:

```
echo 'cGFzc3dvcmQ=' | base64 --decode
```

Encode new value:

```
echo -n 'mypassword' | base64
```

77. Checking Pod Storage Paths

Check mounted volumes:

```
kubectl get pod pod -o jsonpath='{.spec.volumes}'
```

Check volume mount path:

```
kubectl get pod pod -o jsonpath='{.spec.containers[*].volumeMounts}'
```

78. Live Pod Debug Session

```
kubectl debug --image=busybox pod-name
```

This creates a temporary container INSIDE pod for debugging.

79. Upgrade Kubernetes Without Downtime (Cluster Admin)

Check nodes before upgrade:

```
kubectl get nodes
```

Upgrade node one-by-one:

```
kubectl drain node1 --ignore-daemonsets
```

Then update node OS/Kubelet.

```
kubectl uncordon node1
```

80. Advanced kube-proxy Debugging

View kube-proxy rules:

```
iptables -L -t nat
```

Inspect kube-proxy logs:

```
journalctl -u kube-proxy -f
```